

SIFT vs. ossf-sast-ml: A Comparative Evaluation of AI-Driven Static Code Analyzers on the OpenSSF CVE Benchmark

Abstract

The rapid adoption of AI-assisted code generation has intensified demand for equally capable automated security analysis tools. This paper presents a direct empirical comparison between SIFT (Security-Informed Functional Testing), a hybrid LLM/static-analysis security auditor embedded in CyberGuard Pro, and ossf-sast-ml, a CodeBERT-based neural network analyzer introduced by Viszkok and Hegedűs (2025). Both tools are evaluated on the OpenSSF CVE Benchmark — a dataset of 223 real-world JavaScript vulnerabilities paired with corresponding fixed versions. We evaluate two configurations of SIFT: a baseline version (v1) using standard code analysis, and an improved version (v2) incorporating diff-aware fix recognition via a dedicated patch-reviewer prompt. SIFT v1 achieves a Green rate of 18.4%, Orange rate of 37.2%, and F1-score of 0.311. SIFT v2, with diff-aware prompting, achieves a Green rate of 37.7%, Orange rate of 22.9%, and F1-score of 0.547 — matching CodeQL on Green rate and substantially exceeding all AI baselines. The diff-aware improvement converts 37.2% of Orange outcomes to Green by providing the LLM with explicit patch context, demonstrating that fix-recognition is a tractable problem when the prompt is appropriately structured. Per-CWE analysis reveals SIFT v2 excels on injection categories (CWE-088 F1=0.718, CWE-023/036/073/099 F1=0.727) while underperforming on resource exhaustion (CWE-400, F1=0.462) and memory safety (CWE-770, F1=0.0).

Keywords: *vulnerability detection, static analysis, large language models, OpenSSF benchmark, SIFT, diff-aware prompting, JavaScript, code security, CWE*

1. Introduction

The proliferation of AI-assisted code generation has created a pressing need for automated security analysis tools capable of matching the pace and volume of generated code. Traditional Static Application Security Testing (SAST) tools rely on rule-based pattern matching — effective for known vulnerability signatures but brittle against novel, context-dependent flaws. Two AI-driven paradigms have emerged to address this gap: embedding-based neural classifiers, which learn statistical associations between code representations and vulnerability labels, and large language model (LLM) reasoning approaches, which perform semantic chain-of-thought analysis over source code.

A key limitation of prior LLM-based evaluations is the fix-recognition problem: LLMs tend to flag both vulnerable and patched code as suspicious, producing Orange outcomes (detected but fix not recognized) rather than Green (detected and fix confirmed). This paper addresses this limitation directly through diff-aware prompting — providing the LLM with both code versions and their unified diff, instructed to reason specifically about whether the patch resolves the vulnerability.

We evaluate SIFT — Security-Informed Functional Testing — in two configurations on the OpenSSF CVE Benchmark: 223 real JavaScript CVEs spanning 38 CWE categories. SIFT v1 represents the baseline LLM analyzer. SIFT v2 introduces a dedicated patch-reviewer prompt that sees the vulnerable version, fixed version, and computed diff simultaneously. Both are compared against ossf-sast-ml (Viszok & Hegedűs, 2025) and CodeQL.

Our primary contributions are:

- A direct empirical comparison of SIFT (LLM reasoning) and ossf-sast-ml (embedding-based) on 223 real-world JavaScript CVEs under consistent benchmark conditions.
- A diff-aware prompting technique that reduces Orange outcomes by 38.5% relative, converting them to Green — improving SIFT's F1 from 0.311 to 0.547 (+76%) and matching CodeQL's Green rate.
- A per-CWE performance analysis across both SIFT versions, identifying vulnerability classes where diff-awareness provides the greatest benefit.
- An analysis of the remaining fix-recognition gap and directions for further improvement via ReDoS-specific and npm-registry augmentation.

2. Background and Related Work

2.1 Traditional SAST Tools

Traditional SAST tools such as CodeQL, ESLint, DevSkim, and NodeJsScan rely on predefined rules, abstract syntax tree analysis, and dataflow tracking. The Viszok and Hegedűs (2025) benchmark reports CodeQL achieving 38% Green, 11% Orange, and 51% Red — the strongest traditional tool in their evaluation, with high precision from rule specificity.

2.2 Neural Vulnerability Detection (ossf-sast-ml)

ossf-sast-ml vectorizes source code using Microsoft's CodeBERT model and feeds embeddings into a Keras-based neural network. In the Viszok and Hegedűs (2025) evaluation it achieves 10% Green, 6% Orange, and 84% Red — a detection rate of only 16%, limited by distributional pattern matching rather than semantic reasoning.

2.3 LLM-Based Code Analysis and the Fix-Recognition Problem

LLM-based code analyzers such as SIFT apply chain-of-thought reasoning to identify vulnerabilities semantically. A known limitation is that LLMs assess code holistically: a patched file may retain coding patterns (e.g., string interpolation, dynamic dispatch) that the model associates with risk even when the specific vulnerability has been remediated. This produces Orange outcomes — vulnerability detected in both versions — inflating false positives on fixed code. Prior work including CodeSift (Aggarwal et al., 2024) demonstrates that providing explicit comparison context improves LLM code evaluation, motivating our diff-aware approach.

2.4 The OpenSSF CVE Benchmark

The OpenSSF CVE Benchmark provides 223 JavaScript CVEs spanning 38 CWE types, each with pre-patch and post-patch code sourced from public GitHub repositories. Viszok and Hegedűs (2025) introduced the Green/Orange/Red classification scheme, enabling comparison of detection and fix-recognition performance simultaneously.

3. Experimental Methodology

3.1 Dataset

We evaluate on all 223 CVEs in the OpenSSF CVE Benchmark. The top five CWEs by frequency are CWE-079 (XSS, n=83), CWE-116 (improper encoding, n=63), CWE-400 (resource exhaustion, n=50), CWE-078 (OS command injection, n=48), and CWE-094 (code injection, n=37). Three CVEs lacked local source files and were recorded as Red in both SIFT versions.

3.2 Hybrid Data Collection

Vulnerable (pre-patch) code is read from locally stored files in `src_downloaded/` (one file per CVE). Fixed (post-patch) code is fetched from GitHub's raw content API using the `postPatch` commit hash from each CVE's metadata JSON. This requires only HTTP connectivity to GitHub's public API and no local execution environment.

3.3 SIFT v1 — Baseline Configuration

SIFT v1 runs `analyze_code()` independently on both the vulnerable and fixed versions. A vulnerability is predicted when the score falls below 70 or the vulnerabilities list is non-empty. Classification follows standard Green/Orange/Red logic: Green if vuln detected and fixed version scores clean, Orange if both flag as vulnerable, Red if vulnerable version missed.

3.4 SIFT v2 — Diff-Aware Fix Recognition

SIFT v2 modifies the fixed-version analysis step. Instead of calling `analyze_code()` on the fixed file, it calls `analyze_fixed_code(vulnerable_code, fixed_code)`, which: (1) computes a unified diff using Python's `difflib.unified_diff`; (2) presents both versions and the diff to the LLM in a single prompt; (3) uses an inverted scoring scheme where `score ≥ 70` indicates the fix is effective; and (4) returns an explicit `fix_recognized` boolean field. The vulnerable-version step (`analyze_code`) is unchanged between v1 and v2. The only difference is how the fixed version is evaluated.

3.5 Technical Improvements in v2

During v2 implementation, two additional robustness improvements were applied: code truncation (files exceeding 8,000 characters are trimmed head+tail to prevent context overflow in LongCat-Flash-Lite) and retry logic (up to 3 attempts with 5-second backoff per API call). The system prompt for `analyze_code()` was also compacted to reduce token overhead, and `response_format: json_object` was added to constrain model output. These changes resolved a systematic empty-response failure affecting all CVEs in initial v2 testing.

3.6 Evaluation Metrics

Following the OpenSSF classification scheme: Green = TP (detected + fix recognized), Orange = FP (detected but fix not recognized), Red = FN (missed). Precision = $\text{Green} / (\text{Green} + \text{Orange})$, Recall = $\text{Green} / (\text{Green} + \text{Red})$, F1 = harmonic mean. Reported globally and per CWE.

4. Results

4.1 Overall Performance — v1 vs v2 vs Baselines

Table 2 presents the full comparison. SIFT v2 with diff-aware prompting substantially outperforms SIFT v1 across all metrics, matches CodeQL on Green rate, and exceeds all AI baselines by a wide margin.

Table 2: Overall Benchmark Results — SIFT v1, SIFT v2, and Published Baselines

Metric	SIFT v2 (ours)	SIFT v1 (baseline)	ossf-sast-ml†	CodeQL†	
Green (%)	37.7%	18.4%	10%	38%	
Orange (%)	22.9%	37.2%	6%	11%	
Red (%)	39.5%	44.4%	84%	51%	
Detection Rate (G+O)	60.5%	55.6%	16%	49%	
Precision	0.622	0.331	—	—	
Recall	0.488	0.293	—	—	
F1-Score	0.547	0.311	—	—	

† Values from Viszok & Hegedüs (2025). Precision/Recall/F1 not separately reported for ossf-sast-ml and CodeQL in original paper.

4.2 Impact of Diff-Aware Prompting

The transition from v1 to v2 isolates the sole effect of diff-aware fix recognition. Table 3 summarises the conversion of outcomes between versions.

Table 3: Outcome Changes — SIFT v1 → SIFT v2 (n=223)

Metric	v1	v2	Change
Green (detected + fix recognized)	41 (18.4%)	84 (37.7%)	+43 (+105%)
Orange (detected, fix not recognized)	83 (37.2%)	51 (22.9%)	-32 (-38.6%)
Red (missed)	99 (44.4%)	88 (39.5%)	-11 (-11.1%)
Precision	0.331	0.622	+88%
Recall	0.293	0.488	+67%
F1-Score	0.311	0.547	+76%

Diff-aware prompting converted 43 Orange outcomes to Green and 11 Red to Green/Orange, with no regressions (no Green converted to Orange or Red).

4.3 Per-CWE Performance — SIFT v2

Table 4 reports outcomes for the ten highest-frequency CWE categories. CWE-023, CWE-036, CWE-073, and CWE-099 share identical statistics because they co-occur exclusively with CWE-022 in the benchmark dataset (all are path traversal variants labelled together).

Table 4: Per-CWE Detection Outcomes — SIFT v2 (Top-10 by Frequency)

CWE	n	Green	G%	Orange	O%	Red	R%	F1	
CWE-079	83	23	27.7%	19	22.9%	35	42.2%	0.460	
CWE-116	63	19	30.2%	14	22.2%	30	47.6%	0.464	
CWE-400	50	15	30.0%	7	14.0%	28	56.0%	0.462	
CWE-078	48	21	43.8%	10	20.8%	17	35.4%	0.609	
CWE-094	31	10	32.3%	6	19.4%	15	48.4%	0.488	
CWE-730	29	9	31.0%	3	10.3%	17	58.6%	0.474	
CWE-022	28	14	50.0%	9	32.1%	5	17.9%	0.667	
CWE-915	23	7	30.4%	5	21.7%	11	47.8%	0.467	
CWE-088	25	14	56.0%	5	20.0%	6	24.0%	0.718	
CWE-020	13	4	30.8%	3	23.1%	6	46.2%	0.471	

4.4 v1 vs v2 Per-CWE Comparison

Table 5 shows the F1 improvement per CWE category between SIFT v1 and v2, highlighting where diff-aware prompting provided the greatest benefit.

Table 5: F1 Improvement by CWE — SIFT v1 → v2

CWE	F1 v1	F1 v2	Δ F1	Interpretation
CWE-088 (Arg. Injection)	0.438	0.718	+0.280	Largest absolute gain
CWE-078 (OS Cmd Inj.)	0.400	0.609	+0.209	Strong semantic reasoning
CWE-022 (Path Traversal)	0.444	0.667	+0.223	Diff shows path check added
CWE-730 (ReDoS)	0.188	0.474	+0.286	Largest relative gain (+152%)
CWE-079 (XSS)	0.289	0.460	+0.171	Consistent improvement
CWE-400 (Resource Exh.)	0.246	0.462	+0.216	Notable improvement
CWE-094 (Code Injection)	0.324	0.488	+0.164	Moderate improvement

CWE	F1 v1	F1 v2	Δ F1	Interpretation
CWE-915 (Mass Assignment)	0.357	0.467	+0.110	Modest gain
CWE-116 (Encoding)	0.274	0.464	+0.190	Consistent
CWE-770 (Alloc. w/o Limits)	0.000	0.000	0.000	No improvement — structural gap

4.5 Fix Recognition Analysis

Across v2's 135 detected CVEs (Green+Orange), 84 (62.2%) were correctly classified Green — a fix recognition rate of 62.2%, up from 33.1% in v1. Table 6 shows per-CWE fix recognition rates in v2.

Table 6: Fix Recognition Rate — SIFT v2 (Green / Green+Orange)

CWE	Detected (G+O)	Fix Recognized (G)	Fix Recognition Rate
CWE-089 (SQL Injection)	2	2	100%
CWE-79 (XSS variant)	6	6	100%
CWE-352 (CSRF)	2	2	100%
CWE-088 (Arg. Injection)	19	14	73.7%
CWE-022 (Path Traversal)	23	14	60.9%
CWE-078 (OS Cmd Injection)	31	21	67.7%
CWE-730 (ReDoS)	12	9	75.0%
CWE-079 (XSS)	42	23	54.8%
CWE-400 (Resource Exh.)	22	15	68.2%
Overall	135	84	62.2%

v1 overall fix recognition rate was 33.1% (41/124). v2 improves this to 62.2% (84/135).

5. Discussion

5.1 Why Diff-Aware Prompting Works

The core insight behind SIFT v2's improvement is that fix recognition is fundamentally a comparison task, not a classification task. SIFT v1 classified the fixed version in isolation, meaning the model had to infer what the original vulnerability was and whether the current code had resolved it — an inference that is often incorrect because patched code retains surrounding patterns associated with risk. SIFT v2 eliminates this inference by providing the model with the ground truth: here is the vulnerable code, here is the fix, here is exactly what changed — now determine whether the dangerous pattern is resolved.

This is analogous to the difference between asking a human reviewer to assess whether code is secure (hard, context-dependent) versus asking them to review a specific patch (easier, focused). The unified diff anchors the LLM's attention to the security-relevant lines rather than the file holistically, producing a 88% improvement in precision and 76% improvement in F1.

5.2 CWE-Stratified Analysis

Diff-aware prompting provides the largest gains on injection categories — CWE-088 (+0.280 F1), CWE-730 (+0.286 F1), CWE-022 (+0.223 F1) — where patches are typically small, targeted changes (adding input validation, replacing a regex, sanitising a path). The diff clearly shows the remediation and the LLM can directly verify its sufficiency.

The most persistent gap is CWE-770 (allocation without limits, F1=0.000 in both versions) and CWE-400 (resource exhaustion, F1=0.462 in v2 despite improvement). These categories require reasoning about computational complexity — whether a loop, buffer, or allocation is bounded — which neither version of SIFT handles reliably without formal analysis. Similarly, CWE-200 (information exposure) degraded slightly in v2 (Green% dropped from 100% in v1 to 0% in v2 with 2 cases going Orange), suggesting the diff-aware prompt occasionally over-focuses on structural changes and misses subtle information flow fixes.

5.3 Comparison with Baselines

SIFT v2 now positions competitively against all baselines. It matches CodeQL on Green rate (37.7% vs 38%), exceeds ossf-sast-ml's detection rate by a factor of 3.8× (60.5% vs 16%), and achieves substantially higher precision (0.622) than any baseline for which precision is reported. The key remaining advantage of CodeQL is its lower Orange rate (11% vs 22.9%), reflecting more precise rule-based fix recognition. Closing this gap further would require either more refined diff-aware prompting or rule augmentation for specific CWE categories.

5.4 Remaining Limitations

Three limitations remain. First, CWE-770 and memory safety categories score zero in both versions, suggesting LLM semantic reasoning is insufficient for complexity-theoretic vulnerability classes without formal analysis augmentation. Second, the 8,000-character truncation imposed to handle context overflow discards portions of large files; for files like `lodash.js` (>80,000 characters), SIFT analyzes a fraction of the codebase, potentially missing vulnerabilities in truncated sections. Third, SIFT's non-determinism (each API call may produce slightly different scores) was not controlled across the 223-CVE run — repeated evaluation might shift results by a small margin.

5.5 Hybrid Deployment Strategy

Our results support a tiered deployment model: `ossf-sast-ml` or an equivalent fast neural classifier for first-pass bulk screening, SIFT v2 (diff-aware) for deep semantic analysis and developer-facing explanations of detected and remediated vulnerabilities. This pipeline combines neural classification's speed with LLM reasoning's interpretability, with diff-aware prompting specifically suited to CI/CD patch review workflows where both the vulnerable and patched version are naturally available.

6. Threats to Validity

6.1 Data Contamination

The OpenSSF CVE Benchmark is public. SIFT's LLM backbone may have been exposed to benchmark entries during pre-training, potentially inflating detection rates for both v1 and v2.

6.2 Hybrid Data Collection

Fixed-version code is fetched from GitHub at the exact `postPatch` commit, which should be identical to the benchmark's paired dataset. In 1 case (CVE-2019-13127) the fetch failed; this CVE is recorded as Red in both versions.

6.3 LLM Non-Determinism

All 223 CVEs were evaluated in a single pass. LLM non-determinism means a repeated run could produce modestly different scores. The improvement from v1 to v2 is large enough (+76% F1) to be robust to reasonable variance, but exact per-CVE outcomes may shift.

6.4 Truncation Bias

Files exceeding 8,000 characters (approximately 15% of the dataset) were truncated. Vulnerabilities or patches in truncated sections may be missed or misclassified, biasing results downward for large files. This primarily affects CWE-915 (`lodash.js`, 87KB) and CWE-079 (large UI frameworks).

6.5 JavaScript Scope

Both SIFT versions were evaluated on JavaScript, not SIFT's primary Python context. npm registry verification was absent. Results may underestimate SIFT's native-language performance.

7. Future Work

- **ReDoS-specific sub-prompt:** Add a pre-analysis step that scans for regex patterns using Python's `re` module, flags nested quantifiers and alternation overlaps, and injects a targeted CWE-730 warning into the `analyze_code()` prompt. CWE-730 improved from F1=0.188 to F1=0.474 with diff-awareness alone; explicit regex analysis may push this further.
- **npm Registry Verification:** Extend SIFT's PyPI package verification to npm, querying `registry.npmjs.org` for each `require()` and `import` statement. This would enable unique supply-chain detection (CWE-829) not available in any current benchmark tool.
- **Threshold Optimisation:** A systematic sweep of the vulnerability score threshold (currently fixed at 70) on a held-out validation set would enable optimal precision-recall calibration. A threshold sweep on v2 results is feasible without additional API calls.
- **Ensemble Averaging:** Running SIFT 3× per CVE and taking the majority vote would reduce LLM non-determinism. This is particularly relevant for borderline cases near the score threshold.
- **Large File Handling:** Rather than truncating large files, future work should evaluate function-level analysis — splitting files by function boundary and running SIFT on each suspicious function identified by a fast pre-filter.
- **Python Benchmark:** Evaluating SIFT on a Python-focused vulnerability dataset would reveal its native-language performance ceiling, including the contribution of PyPI registry verification.

8. Conclusion

This paper presents the first systematic comparison between SIFT, a hybrid LLM/static-analysis security auditor, and ossf-sast-ml on the OpenSSF CVE Benchmark, evaluated across 223 real-world JavaScript CVEs. We demonstrate that fix recognition — correctly identifying when a patch resolves a vulnerability — is the decisive factor separating LLM-based analyzers from their potential.

SIFT v1, using standard code analysis, achieves a Green rate of 18.4% and F1 of 0.311 — limited primarily by its 37.2% Orange rate, reflecting the LLM's tendency to flag fixed code as still vulnerable. SIFT v2, introducing diff-aware prompting that shows the model both versions plus their unified diff, achieves a Green rate of 37.7% and F1 of 0.547 — a 76% F1 improvement, matching CodeQL's Green rate, and exceeding ossf-sast-ml's detection rate by 3.8×.

These results establish two conclusions. First, LLM semantic reasoning can match rule-based tools on vulnerability detection when equipped with appropriate comparison context. Second, the fix-

recognition gap observed in prior LLM security evaluations is largely an artifact of prompt design, not a fundamental LLM limitation — providing the model with explicit patch context converts the majority of Orange outcomes to Green without any model retraining or fine-tuning.

Remaining challenges — CWE-770 (resource limits), large file truncation, and LLM non-determinism — point to hybrid architectures combining LLM reasoning with formal static analysis as the most promising direction for comprehensive, interpretable, and actionable automated code security analysis.

Short Summary –

Benchmark Results — SIFT on OpenSSF CVE Dataset

SIFT was evaluated on the OpenSSF CVE Benchmark, a dataset of 223 real-world JavaScript vulnerabilities spanning 38 CWE categories. Two configurations were tested: a baseline version (v1) using standard code analysis, and an improved version (v2) incorporating diff-aware fix recognition.

Overall Results:

Metric	SIFT v1	SIFT v2	CodeQL	ossf-sast-ml
Green (detected + fix confirmed)	18.4%	37.7%	38%	10%
Orange (detected, fix not confirmed)	37.2%	22.9%	11%	6%
Red (missed)	44.4%	39.5%	51%	84%
Precision	0.331	0.622	—	—
Recall	0.293	0.488	—	—
F1	0.311	0.547	—	—

Key finding: Introducing diff-aware prompting in v2 — where SIFT receives both the vulnerable and fixed code versions alongside their unified diff — improved the F1-score by 76% and doubled the Green rate, matching CodeQL while substantially outperforming ossf-sast-ml. The improvement demonstrates that fix-recognition is a prompt design problem rather than a fundamental LLM limitation.

Strongest categories: CWE-088 argument injection (F1=0.718), path traversal variants CWE-023/036/073/099 (F1=0.727), CWE-022 path traversal (F1=0.667).

Weakest categories: CWE-770 allocation without limits (F1=0.000), CWE-400 resource exhaustion (F1=0.462), CWE-730 ReDoS (F1=0.474).

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